

JKBOSE

2026 CLASS 11TH

Motion in
1D ✓

MOTION IN STRAIGHT LINE

LECTURE-1

PHYSICS

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Topics To Be Covered

- 1** Revision
- 2** Motion
- 3** Reference Point
- 4** Distance/Displacement
- 5** Speed/Velocity
- 6** Acceleration

STOP FOOLING YOURSELF,
YOU KNOW YOU CAN DO BETTER

Recap From 1st Lec.

Maths

- Maths Bohat Aasaan hai
- number
- $+$ $-$ $\times$ $\div$
- Fraction
- Angle / Power
- Trigonometry
- Graph / Equation
- $\frac{dy}{dx}$ $\int y dx$

Fraction

- $\frac{1}{2} = 0.5$
- $\frac{1}{3} = 0.33$
- $\frac{1}{4} = 0.25$
- $\frac{1}{10} = 0.1$
- $\frac{1}{100} = 0.01$
- $5.42 \times 100 = 542$

Power

- $2^2 = 2 \times 2$
- $x^0 = 1$
- $x^a \cdot x^b = x^{a+b}$
- $(x^a)^b = x^{a \times b}$

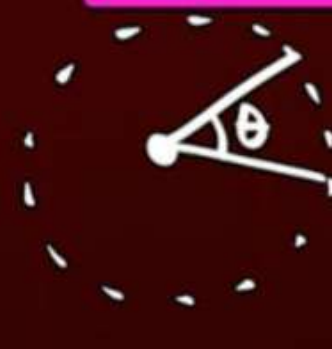
$$\frac{1}{\infty} = 0$$

$$\sqrt{2} = 2^{\frac{1}{2}}$$

$$\sqrt{2} = 1.4$$

$$\sqrt[3]{3} = 1.7$$

Angle θ



$$\theta = \frac{\text{Arc}}{\text{Radius}}$$

0 Rad.

$$\rightsquigarrow 180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$

Recap From 2nd Lec.

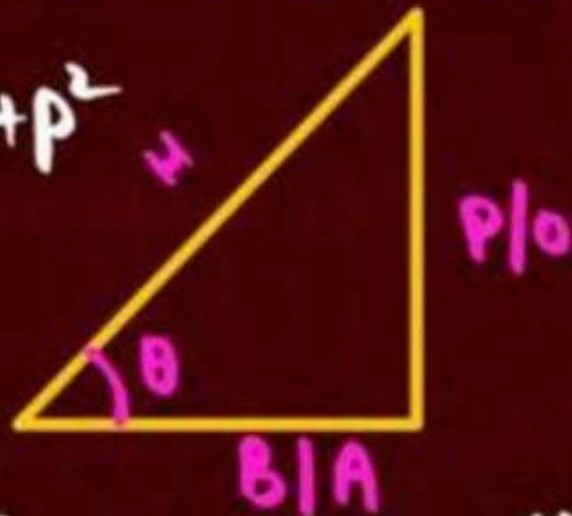
Trigonometry Table

	0°	30°	45°	60°	90°	120°	135°	150°	180°
Angle	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan \theta$	$\frac{0}{1}=0$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\frac{1}{0}=\infty$	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

Trigonometry

Special Δ

$H^2 = B^2 + P^2$



$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$

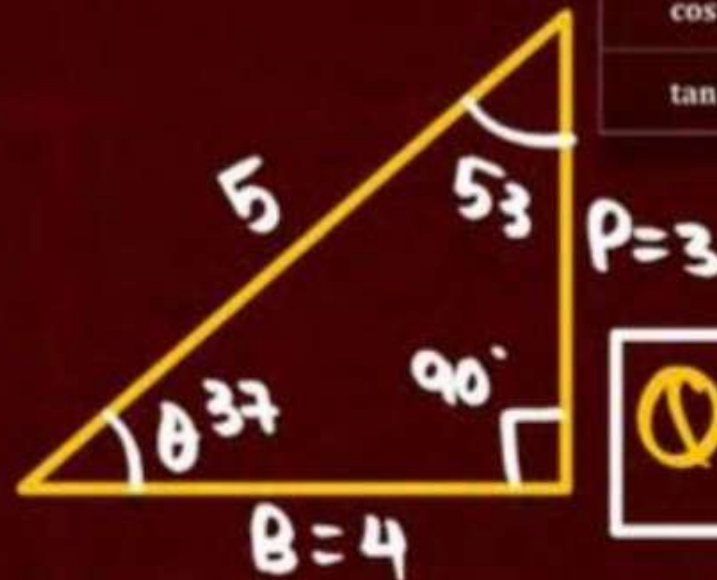
$\sec \theta = \frac{1}{\cos \theta}$

$\cot \theta = \frac{1}{\tan \theta}$

$\sin^2 \theta + \cos^2 \theta = 1$

$\sin \theta = \frac{P}{H}$
 $\cos \theta = \frac{B}{H}$
 $\tan \theta = \frac{P}{B}$

$\operatorname{cosec} \theta = \frac{H}{P}$
 $\sec \theta = \frac{H}{B}$
 $\cot \theta = \frac{B}{P}$



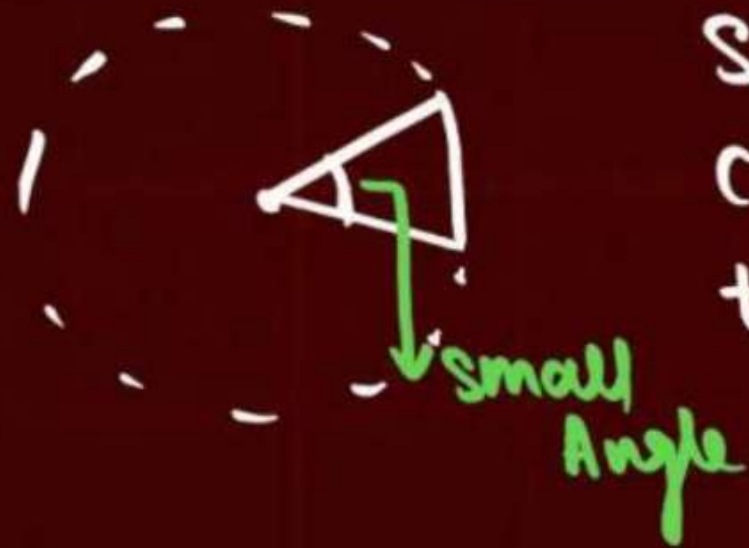
Quadrant System



Recap From 3rd Lec.



Small Angle Approx.



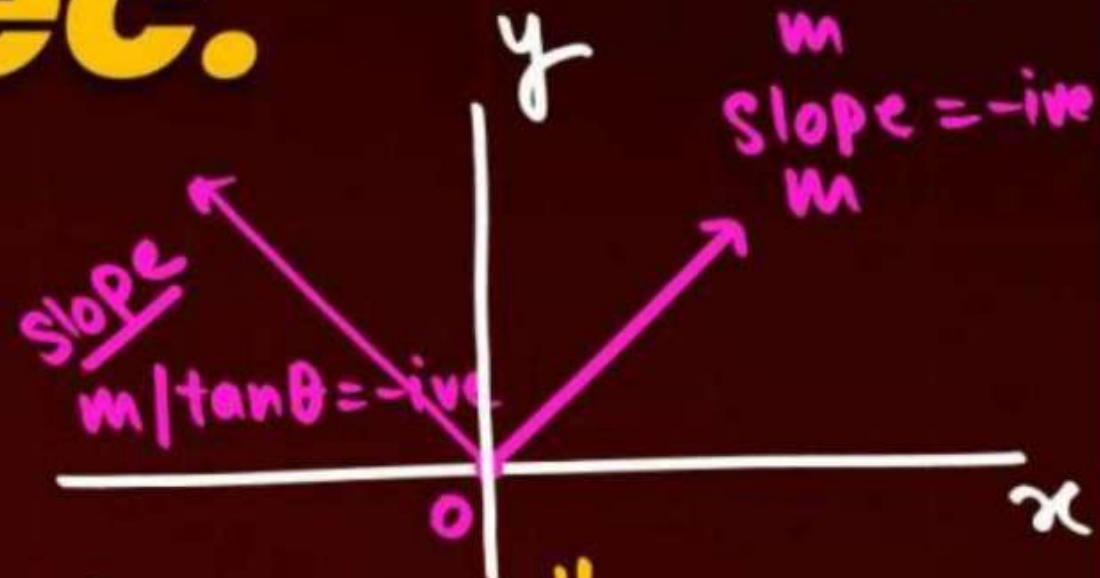
$$\begin{aligned}\sin \theta &= \theta \\ \cos \theta &= 1 \\ \tan \theta &= \theta\end{aligned}$$

Equation

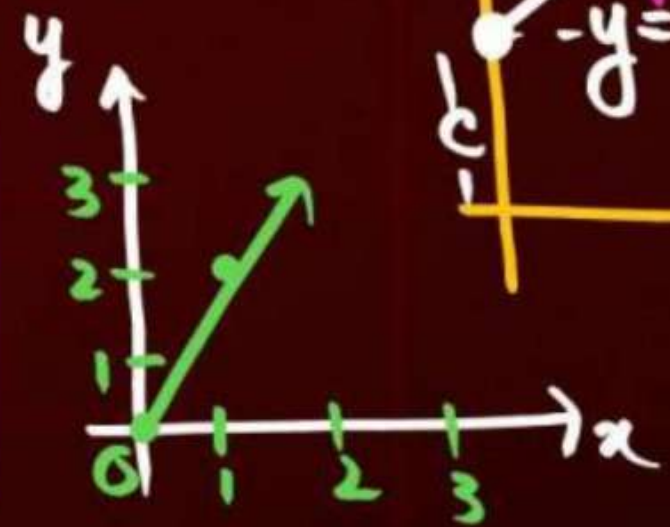
English Statement

Slope $\rightarrow$ $y = \frac{y}{x}x = x \cdot \frac{y}{x}$

x	0	1	2
y	0	2	4



(i) Straight line equation



$$y = mx + c$$

Recap From 4th Lec. 3rd



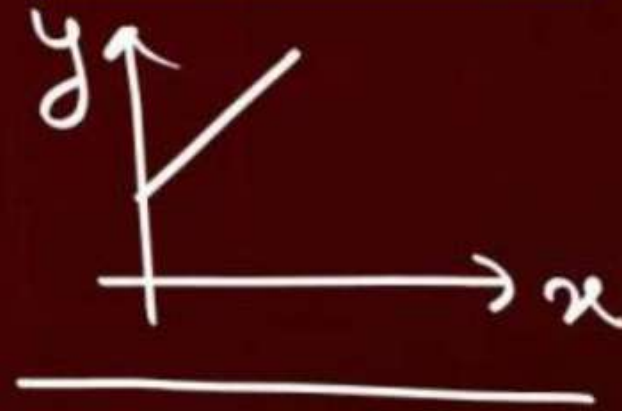
Equation

$$\underline{y = mx + c}$$

$$y = x^2$$

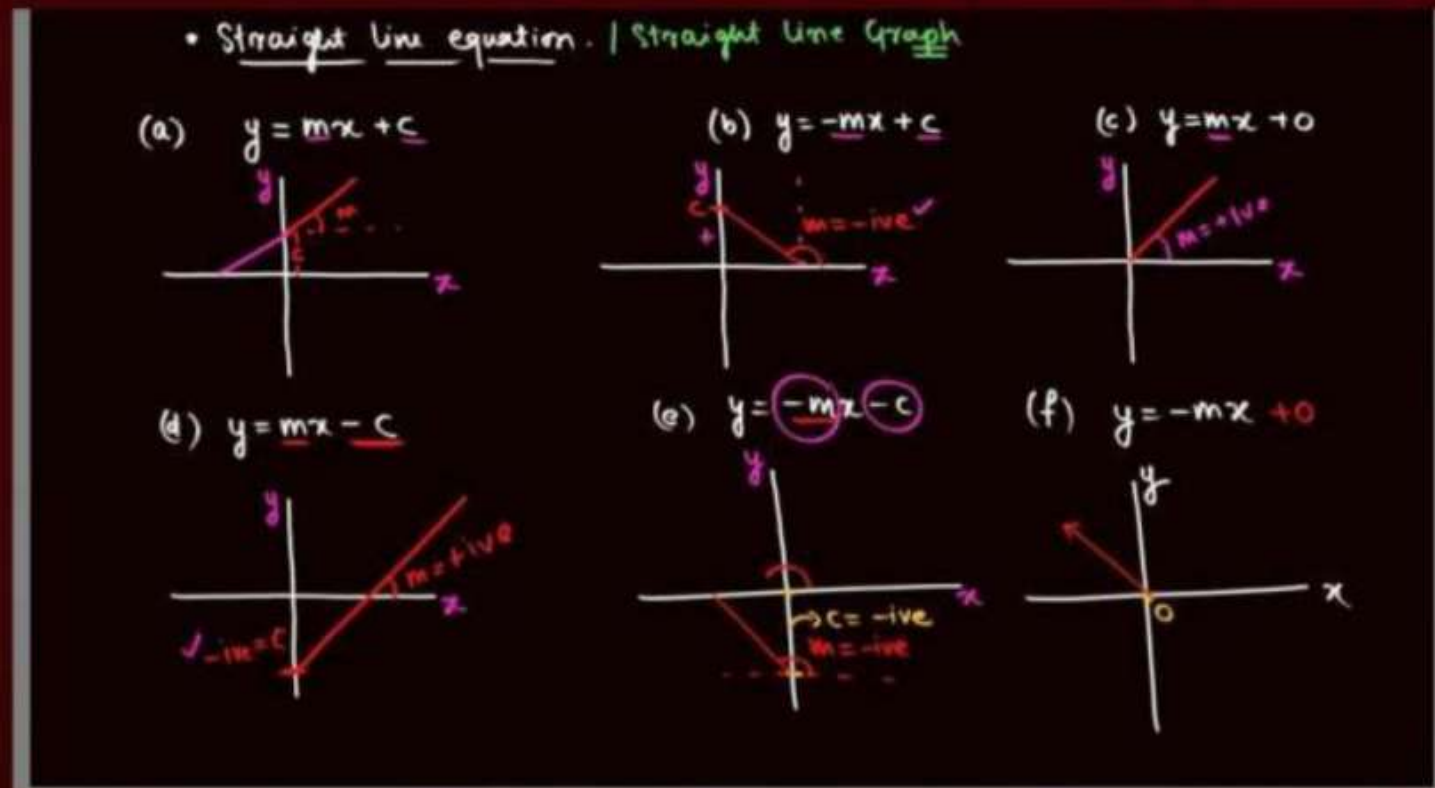
→ Table →

Graph



Recap From 5th Lec.

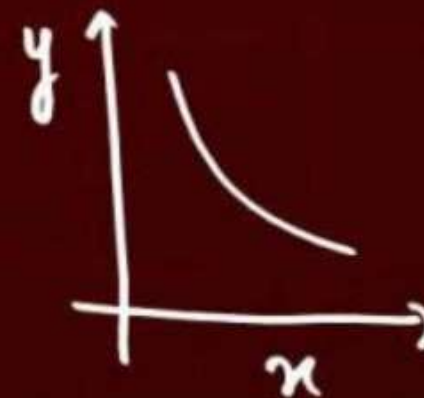
① $y = \underline{mx} + c$ (Important)



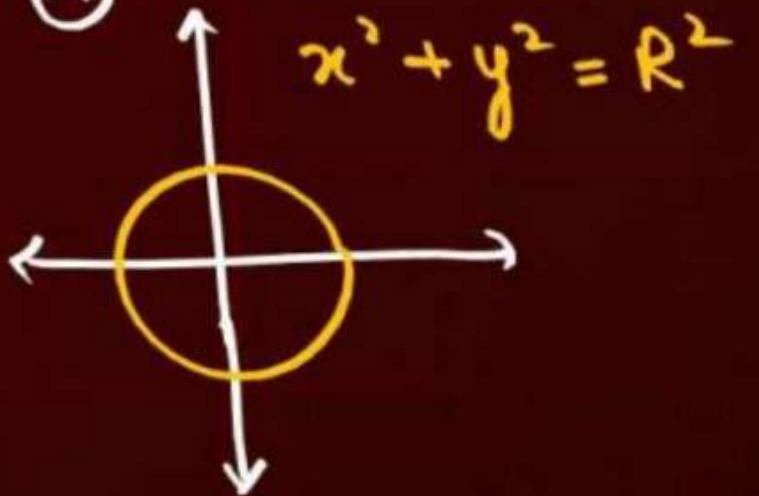
② $y = x^2$



③ $y = \frac{1}{x}$



④ $x^2 + y^2 = R^2$



Recap From 6th Lec.

Differentiation

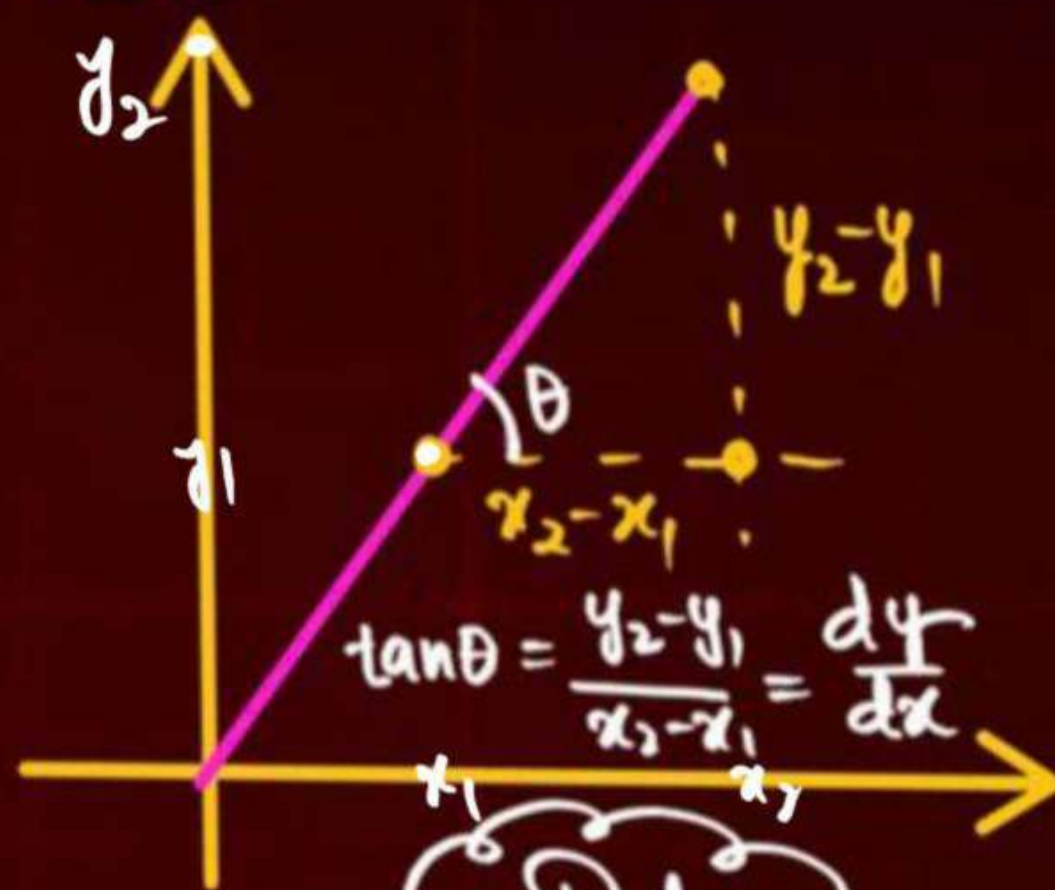
$$y = 2x$$

	x_1	x_2		
x	1	2	3	4
y	2	4	6	8
	y_1	y_2		

$$y = 2x$$

↓
Slope

differentiation $\left(\frac{dy}{dx}\right)$



$$\text{differentiation} = \frac{\text{Small change in } y}{\text{Small change in } x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 2}{2 - 1} = \frac{2}{1} = 2$$

Rules

Recap from lecture 7.

Rules for differentiation

$$\textcircled{1} \checkmark \frac{dc}{dx} = 0$$

$$\textcircled{2} \checkmark \frac{dx^n}{dx} = nx^{n-1}$$

$$\textcircled{3} \checkmark \frac{de^x}{dx} = e^x$$

$$\textcircled{4} \checkmark \frac{d \ln x}{dx} = \frac{1}{x}$$

$$\textcircled{5} \checkmark \frac{d \sin x}{dx} = \cos x$$

$$\textcircled{6} \checkmark \frac{d \cos x}{dx} = -\sin x$$

$$\textcircled{7} \checkmark \frac{d \tan x}{dx} = \sec^2 x$$

$$\textcircled{8} \checkmark \frac{d \csc x}{dx} = -\csc x \cot x$$

$$\textcircled{9} \checkmark \frac{d \sec x}{dx} = \sec x \tan x$$

$$\textcircled{10} \checkmark \frac{d \cot x}{dx} = -\operatorname{cosec}^2 x$$

$$\textcircled{11} \checkmark \frac{d(u+v+w)}{dx} = \frac{du}{dx} + \frac{dv}{dx} + \frac{dw}{dx}$$

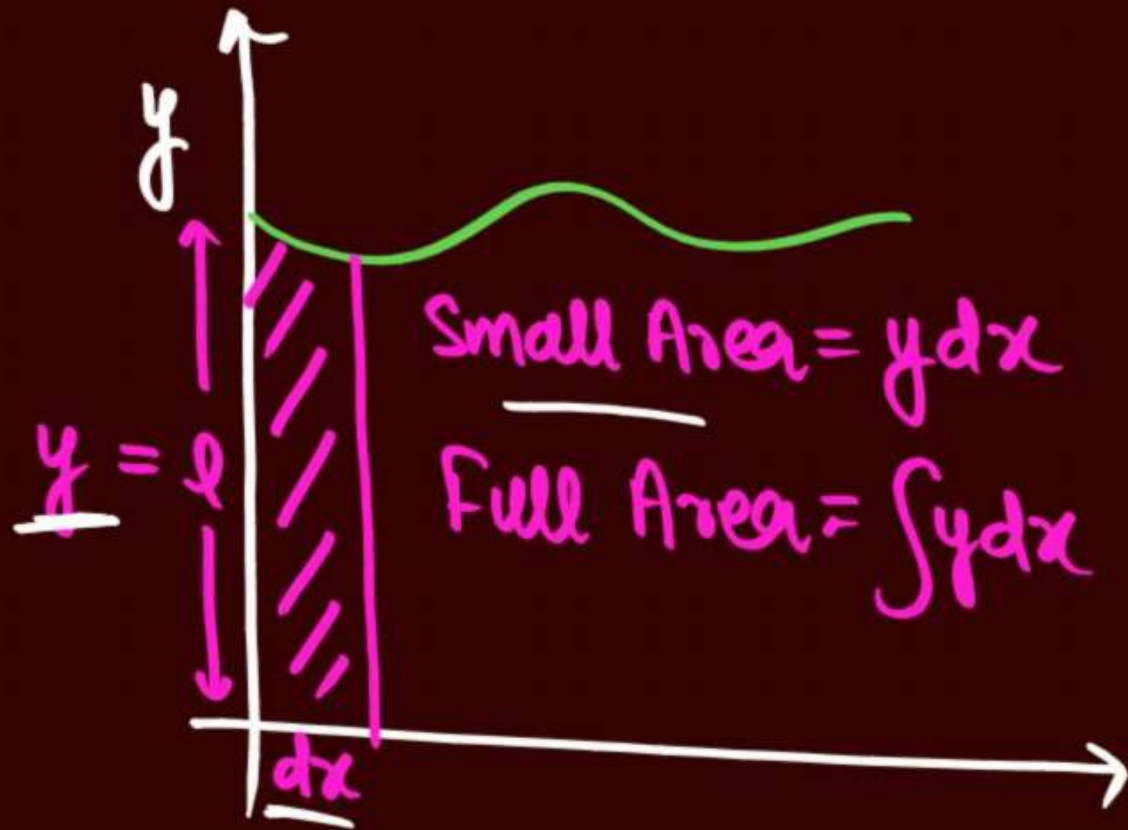
$$\textcircled{12} \checkmark \frac{d(u \cdot v)}{dx} = v \frac{dv}{dx} + u \frac{du}{dx}$$

$$\textcircled{13} \checkmark \frac{d \frac{u}{v}}{dx} = \frac{v \frac{dv}{dx} - u \frac{du}{dx}}{v^2}$$

Recap lecture no. 8.

Integration

- Jorna
- Area under graph



Rule - ①

① ✓ $\int x^n dx = \frac{x^{n+1}}{n+1}$

② ✓ $\int \sin x dx = -\cos x$

③ $\int (u+v+w) dx = \int u dx + \int v dx + \int w dx$

④ $\int (u-v-w) dx = \int u dx - \int v dx - \int w dx$

⑤ $\int (u+v-w) dx = \int u dx + \int v dx - \int w dx$

Chain Rule $\left(\frac{dy}{dx}\right) = \frac{d \sin(x^2)}{dx} = 2x^{2-1} \cos x^2 = 2x^1 \cos x^2$

Integration

$\int y dx$ $\int_{x_1}^{x_2} y dx$

qth

JKBOSE syllabus

Neet → ① x 4 = 4

$\frac{dy}{dx}$

Motion in a Straight Line

Frame of reference, Motion in a straight line, Elementary concepts of differentiation and integration for describing motion, uniform and non-uniform motion, and instantaneous velocity, uniformly accelerated motion, velocity - time and position-time graphs. Relations for uniformly accelerated motion (graphical treatment).

Sydx

Motion

lecture.no. ①



Birds Fly



Fish Swim

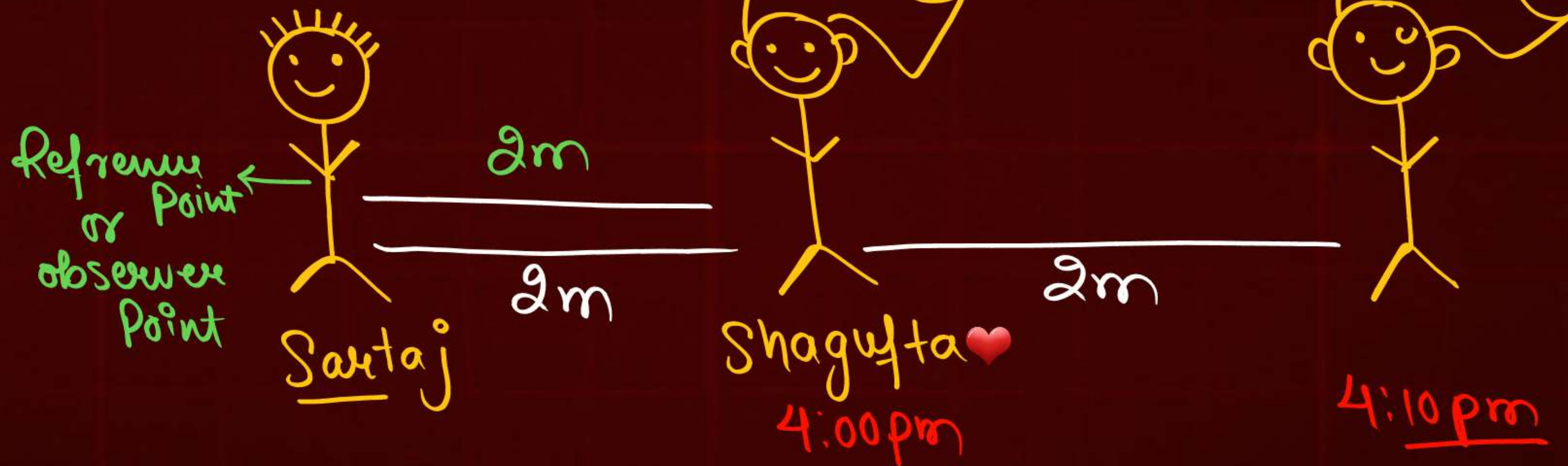


Car is Moving

Motion



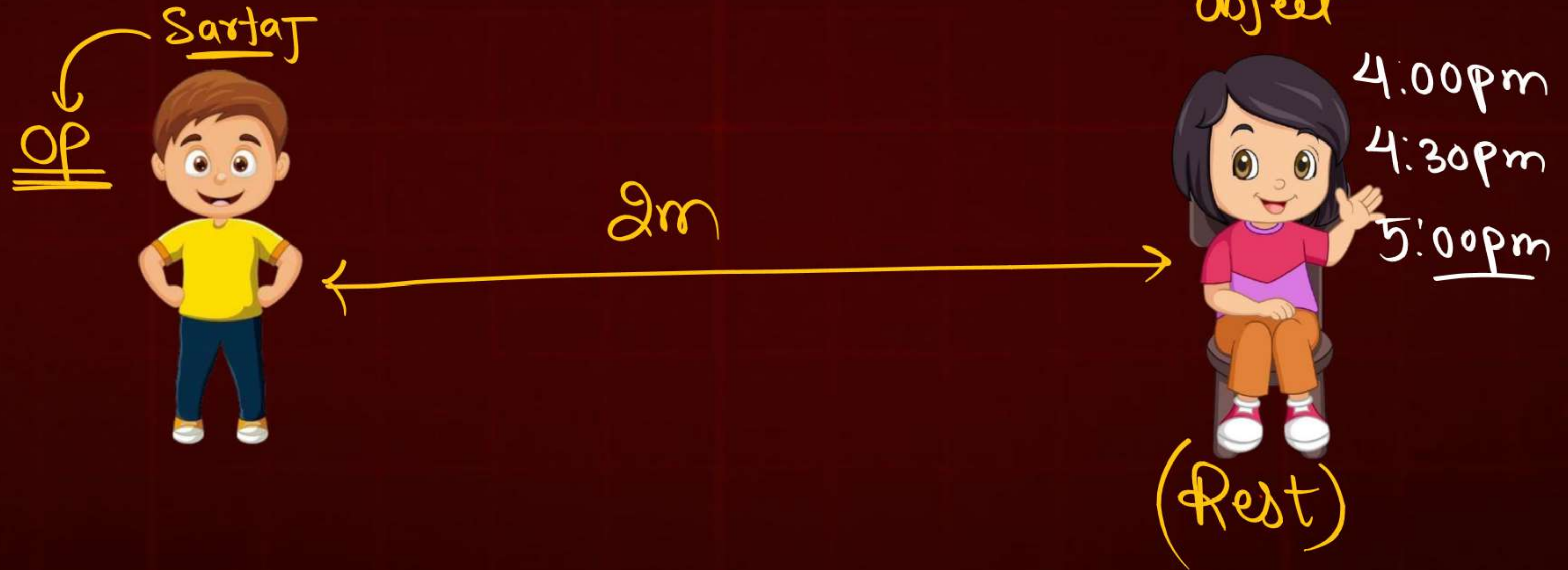
When body/object change its position with respect to (Time) or (Reference point / observer point)



Rest

Position of object does not change with Refrence point / observer point / Time

Example:

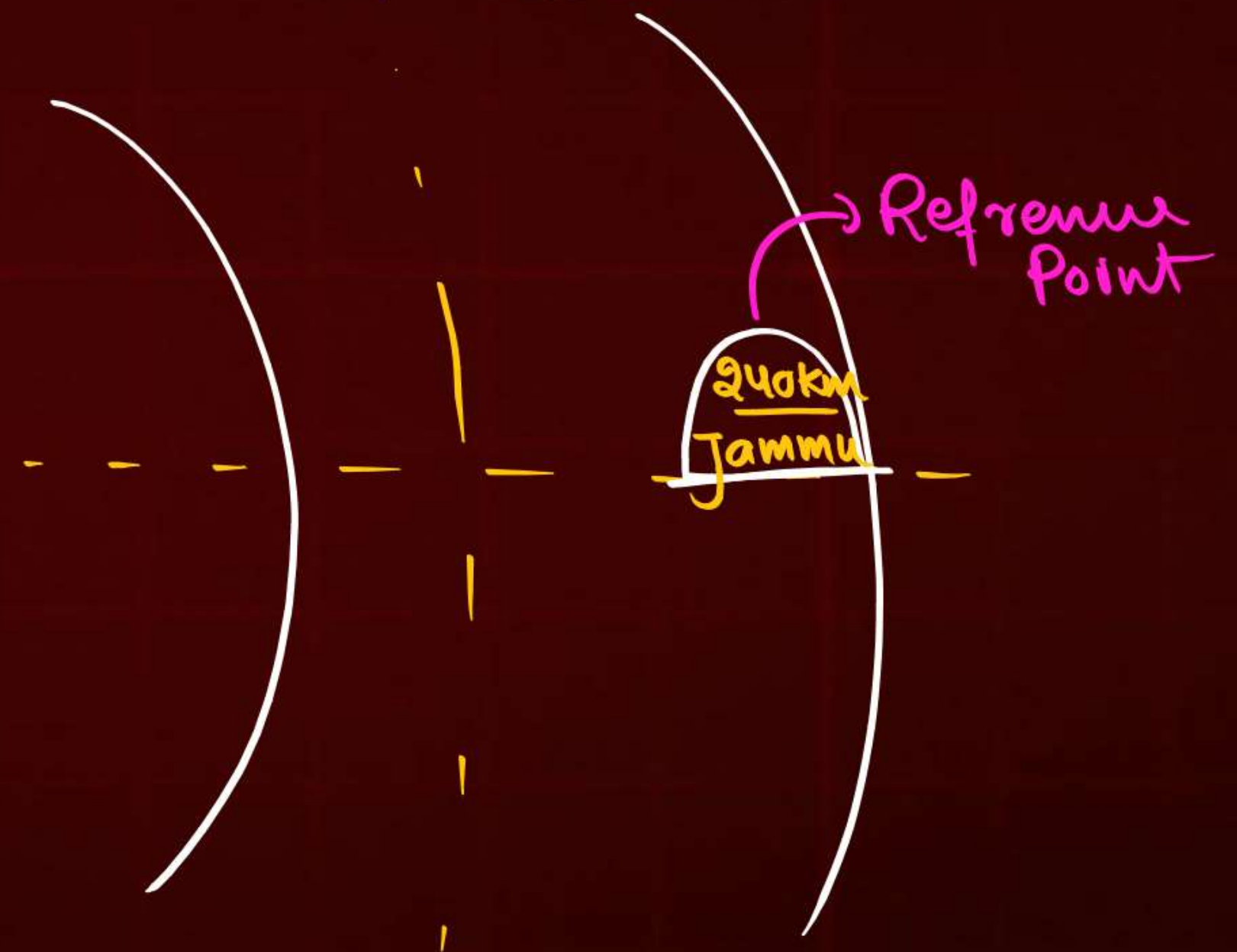


Reference Point

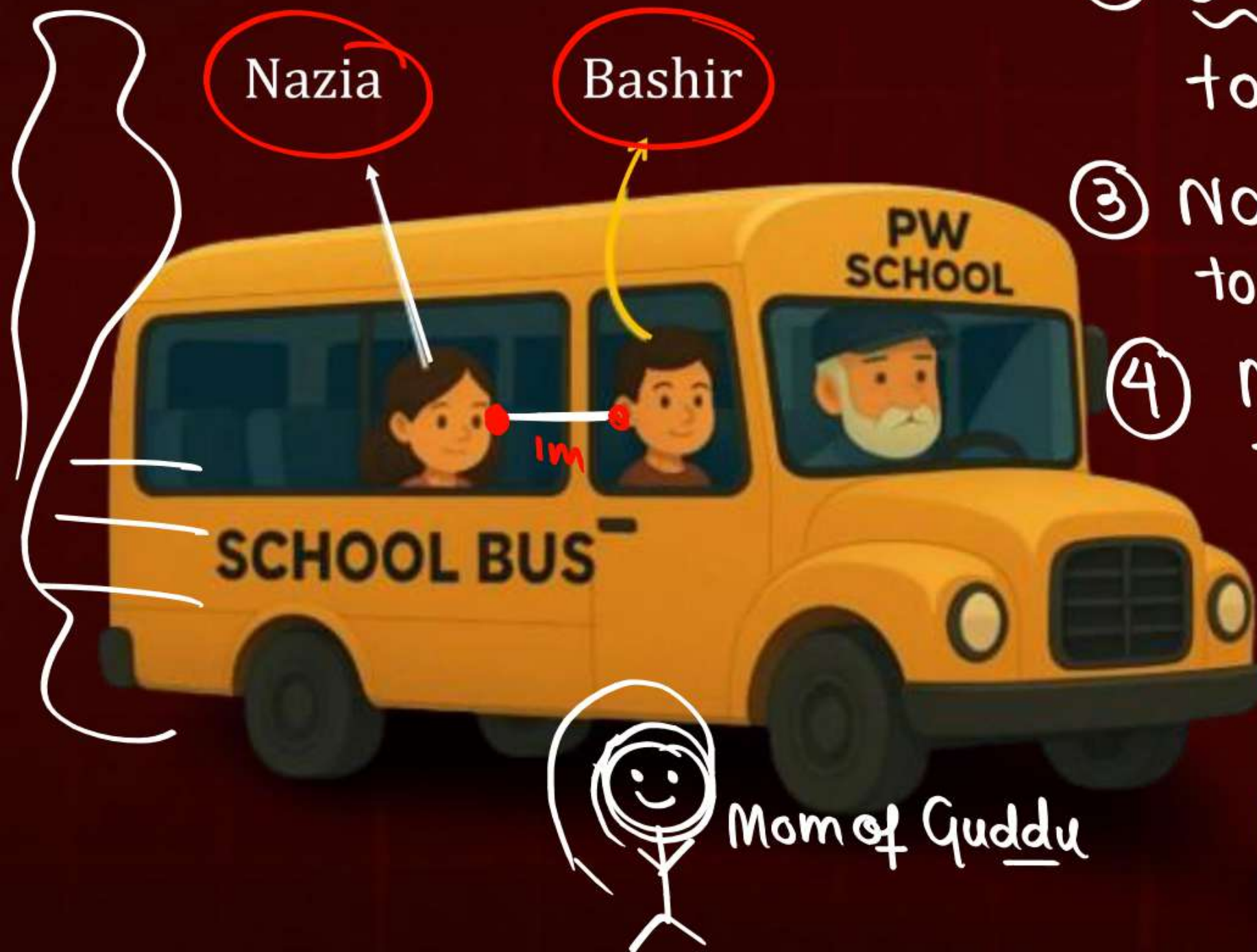


The observer point from where you can see or observe the object

shagufta



Motion is Relative



① Bashir is in Motion with respect to Guddu

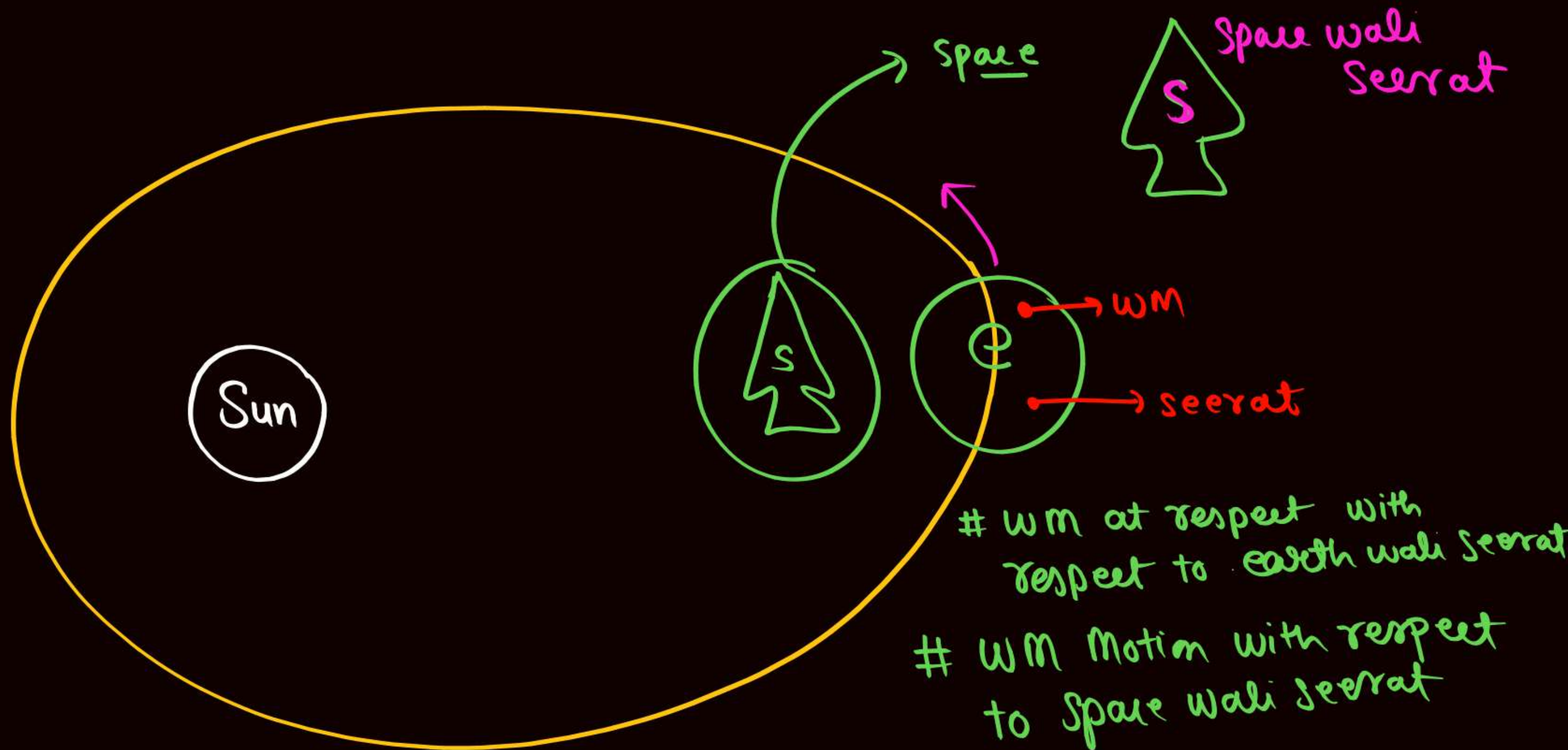
② Bashir is in Rest with respect to Nazia

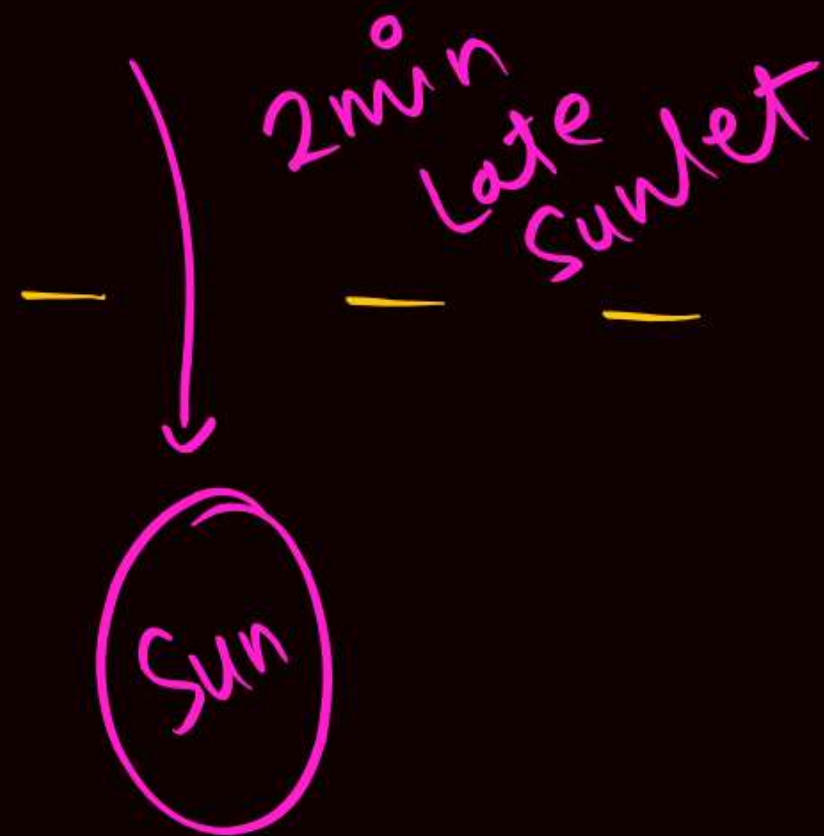
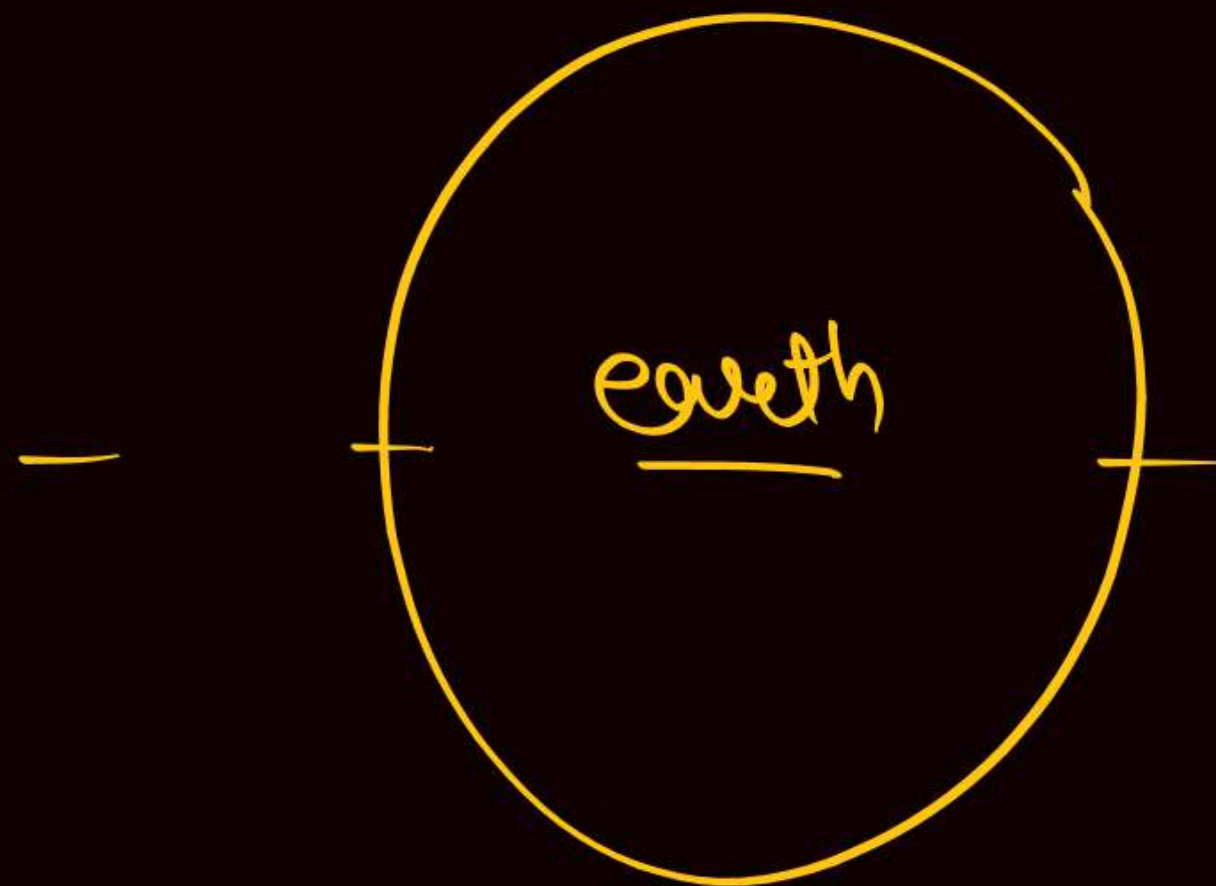
③ Nazia is in Motion with respect to Guddu

④ Nazia is in Rest with respect to Bashir

⑤ Guddu rest with respect to Guddu Mo

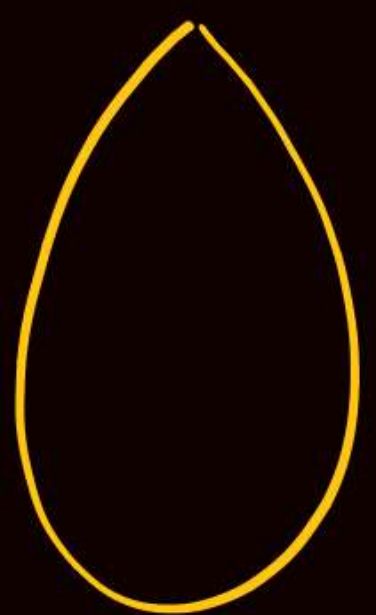
⑥ Guddu is in Motion with respect to N & B



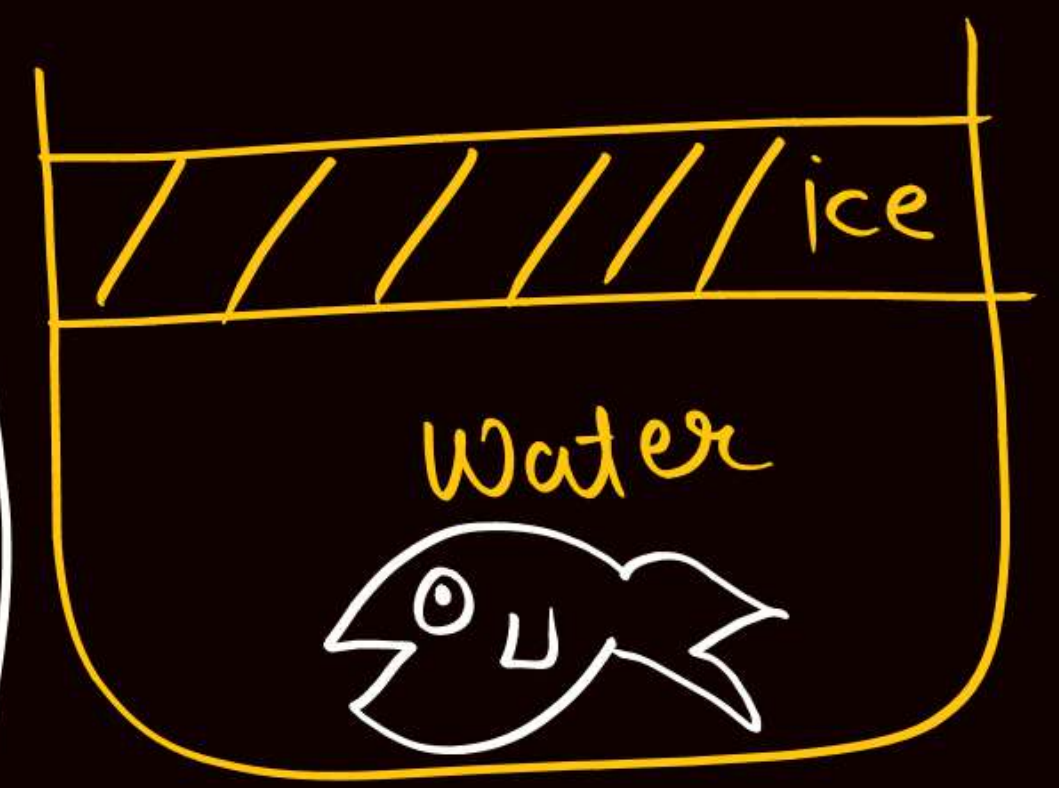
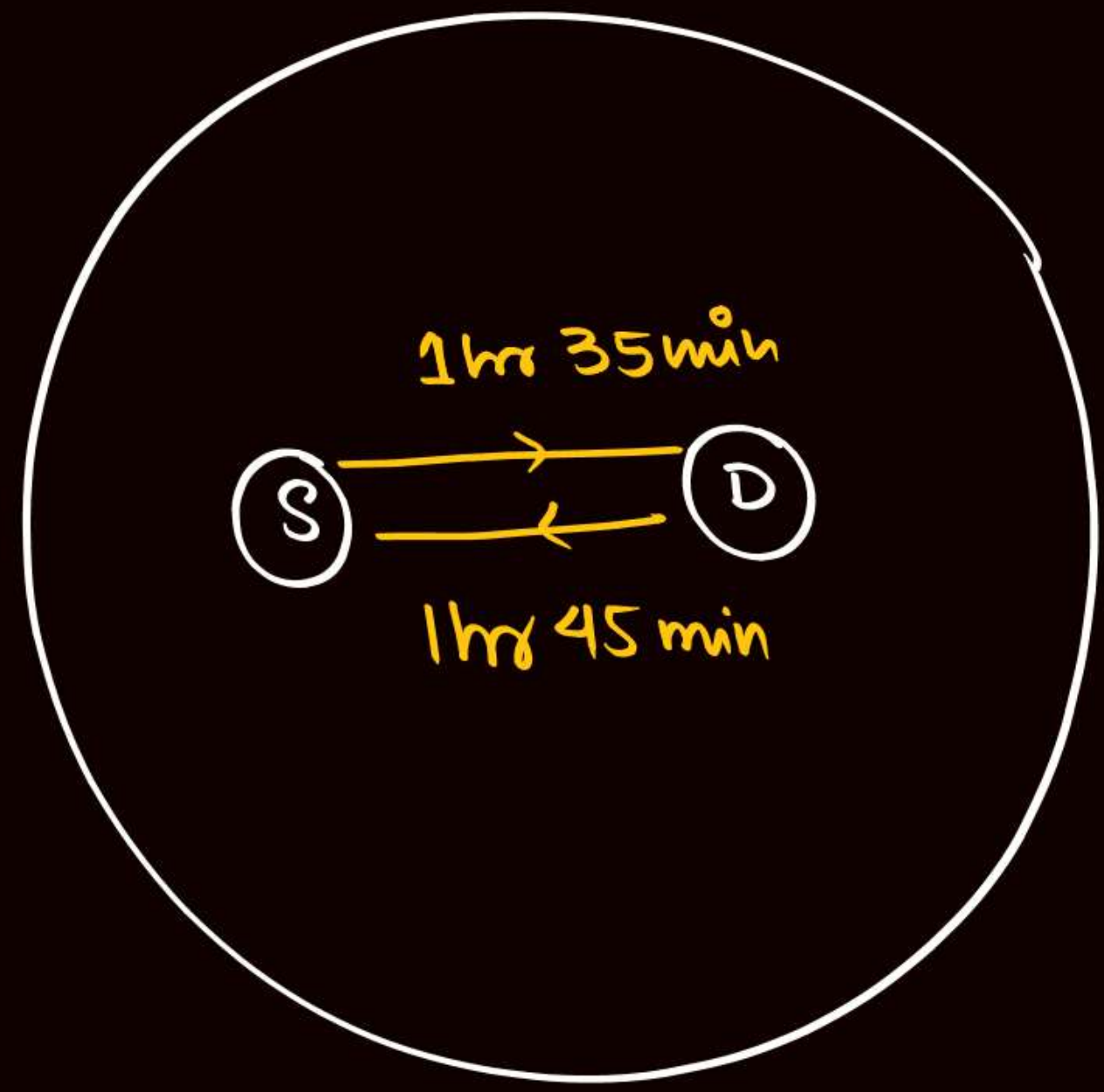
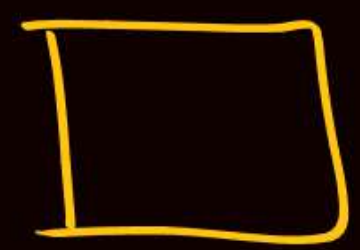


9.

Dal Lake



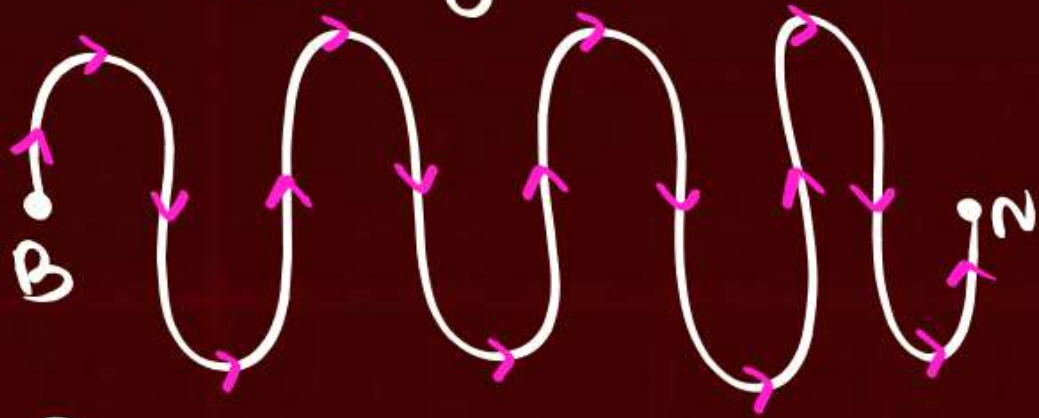
Spherical



Distance

($\mathcal{L}$)

- ① Total Path travel by the Body



- ② Scalar

$$1\text{ km} = 1000\text{ m}$$

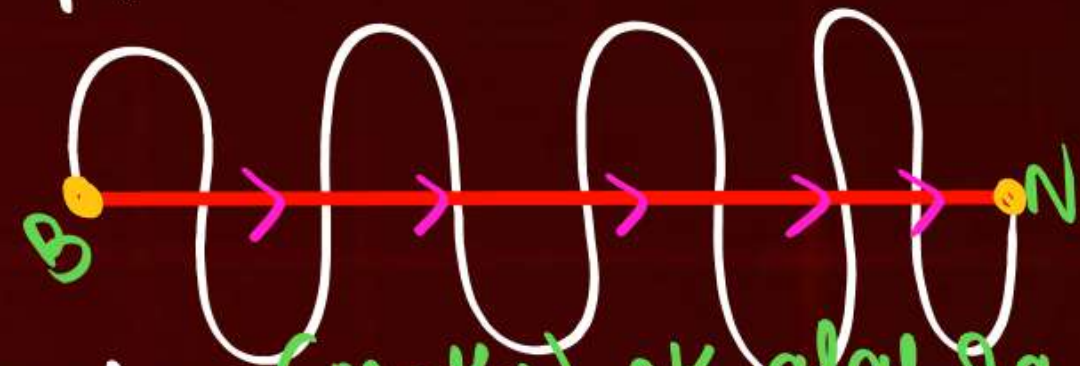
- ③ unit $\rightarrow$ metre (m)

- ④ Positive

$\vec{\mathcal{L}}$

Displacement

- ① Shortest Straight line path travel by the Body is called Displacement



- ② Vector (maths) ek alag sa chapter
Ayee gaar.

- ③ metre

- ④ Zero, -ive, +ive

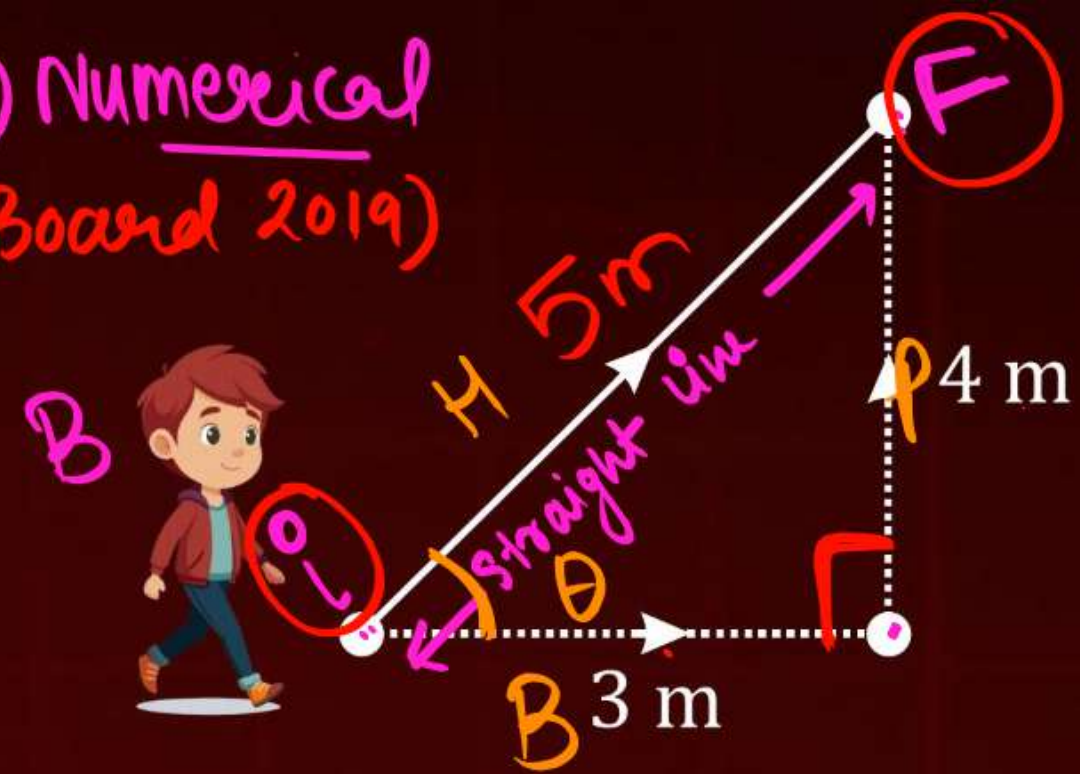
Q1 (Numerical)



? Distance: $5 + 10 + 10 = 25\text{m}$

? Displacement: 5m

Q2 Numerical (Board 2019)



Distance: $3 + 4 = 7\text{m}$

Displacement: 5m

$$H^2 = B^2 + P^2$$

$$H^2 = 3^2 + 4^2$$

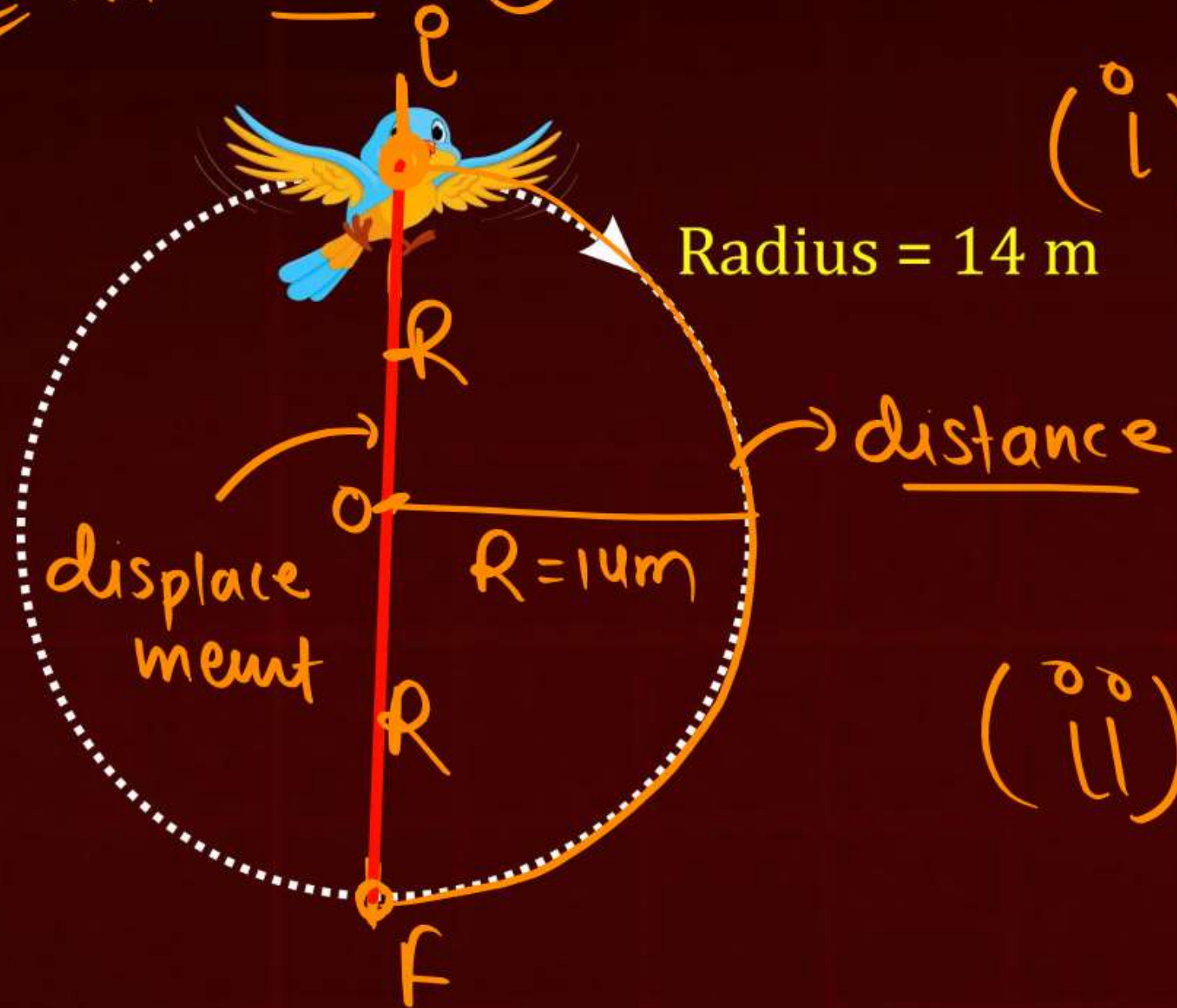
$$H^2 = 9 + 16$$

$$H^2 = 25$$

$$H^2 = 5^2$$

$$H = \sqrt{5^2} = 5\text{m} \checkmark$$

Q Numerical - (3)



After half revolution, find Distance
Displacement

$$(i) \text{ distance} = \frac{\text{Circumference of circle}}{2}$$

$$= \frac{2\pi r}{2} = \pi r$$
$$= \frac{22}{7} \times 14^2$$
$$= 44\text{m}$$

$$(ii) \text{ Displacement}$$
$$= 2R$$
$$= 2 \times 14$$
$$= \underline{+28\text{m}}$$

Numerical - 4



Radius = 14 m

Full revolution, find Distance
Displacement

Sol Distance = Circumference of Circle

$$= 2\pi r$$

$$= 2 \times \frac{22}{7} \times 14$$

$$= 4 \times 22 = \underline{88\text{m}}$$

Displacement

if initial and final point is same

$$\text{displacement} = \underline{0}$$

Q.5 An object has moved through a distance. Can it have zero displacement? If yes, support your answer with an Example

Sol

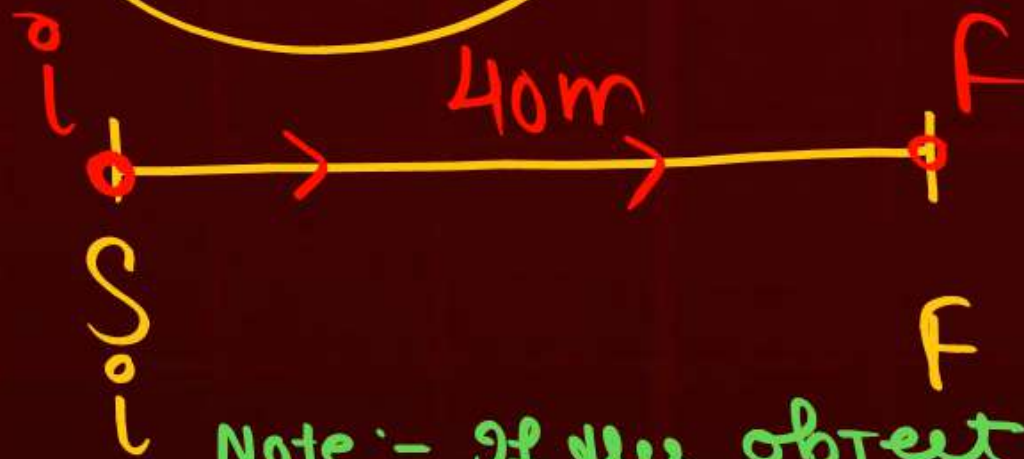


$$\text{Distance} = 2 + 2 = 4\text{m}$$

$$\text{Displacement} = 0 \checkmark$$

Q.6 Can the magnitude of displacement be equal to distance travelled by object?

Sol



$$\begin{aligned} \text{distance} &= 40\text{m} \\ \text{displacement} &= 40\text{m} \end{aligned} \checkmark$$

Note:- If the object travel straight line without turning.

Q7



distance = 10m

✓ displacement = -10m

Speed

(v)

- Rate of change of distance
÷ time

$$\text{Speed} = \frac{\text{distance}}{t}$$

$$v = \frac{x}{t}$$

- Scalar

- unit $v = \frac{x}{t} = \frac{m}{s}$

- Positive

Velocity

$\vec{v}$

- Rate of change of displacement
is velocity

$$\text{Velocity} = \frac{\text{displacement}}{\text{time}}$$

$$\vec{v} = \frac{\vec{x}}{t}$$

- vector

• unit $\vec{v} = \frac{\vec{x}}{t} = \frac{m}{s}$

- Positive / negative / zero

Aane
wale
dino
main

Speed

Average
Speed

Instantaneous
Speed

Velocity

Average
velocity

Instantaneous
velocity

Question

✓ Convert 1 km/hr into m/s

$$1 \text{ km} / \text{hr}$$

- 1 km = 1000 m

- 1 hr = 60 min

$$1 \text{ hr} = 60 \times \underline{1 \text{ min}}$$

$$1 \text{ hr} = \underline{60} \times \underline{60} \text{ sec}$$

$$1 \text{ hr} = \underline{3600 \text{ sec}}$$

$$1 \text{ km/hr} = \frac{1 \text{ km}}{\text{hr}}$$

$$1 \text{ km/hr} = \frac{1 \times 1000 \text{ m}}{3600 \text{ sec}}$$

$$1 \text{ km/hr} = \frac{5}{18} \text{ m/s}$$

$$1 \text{ km/hr} = \frac{5}{18} \text{ m/s}$$

Question

Convert 54 km/h into m/s

54 km/hr ✓

car

$$54 \text{ km/hr} = \text{---} \text{ m/s}$$

$$54 \text{ km/hr} = 54 \times \left(\frac{1 \text{ km}}{\text{hr}} \right) \frac{5 \text{ m/s}}{18}$$

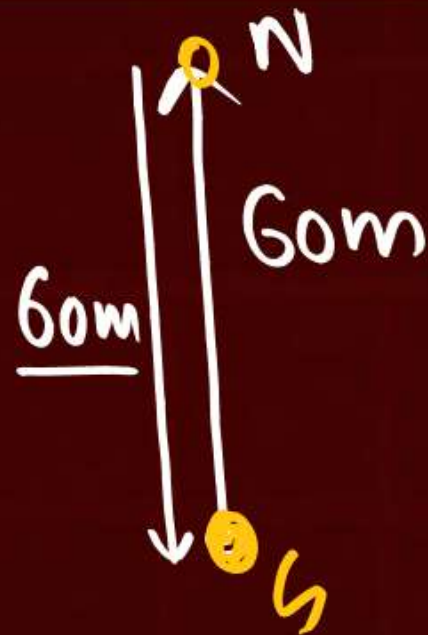
$$= \cancel{54}^3 \times \frac{5}{\cancel{18}} \text{ m/s}$$

$$= \textcircled{15 \text{ m/s}} \text{ ✓ Physics}$$

Chocolate Question

A car moves 60 km towards north and then 60 km towards south in 2 hours. What is its average velocity?

- 1 60 km/h
- 2 30 km/h
- 3 0 km/h
- 4 120 km/h



$$\text{displacement} = 0$$

$$\text{time} = 2 \text{ hr}$$

$$\text{Velocity} = \frac{\text{displacement}}{\text{time}}$$

$$= \frac{0}{2} = 0$$

Chocolate Question

Which of the following statements is correct?

$$\text{Velocity} = \frac{2\text{m}}{\text{s}}$$

- 1 Speed is a vector quantity ✓
- 2 Velocity is a scalar quantity ✓
- 3 Speed has direction ✓
- 4 Velocity has both magnitude and direction ✓

Question

Convert 72 km/h into m/s

Neet = ?

SUMMARY



- ✓ Motion
- ✓ Rest
- ✓ Reference point
- ✓ mR
- ✓ $x / \vec{x}$
- ✓ $v / \vec{v}$
- ✓ numericals

Thank
You